

File Classifier (By NER)

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The file structure expected by the algorithm

The algorithm provides a computationally costly solution to the problem of classifying unsorted files into groups. **The algorithm does not require that filenames have embedded dates.** But, the algorithm does expect to find files in the following directory structure;

1. a folder Y containing m subfolders (Z), each subfolder containing a variable set of documents that talk about the same topic Z;
2. a folder X containing a list of n unsorted documents each one of which needs to be placed in one of the m Z subfolders.

Graphically, this is what the file structure would look like:

Name	Date modified	Type	Size
 3	2/18/2020 10:14 PM	File folder	
 4	2/18/2020 10:15 PM	File folder	
 5	2/18/2020 10:15 PM	File folder	
 6	2/18/2020 10:15 PM	File folder	
 7	2/18/2020 10:15 PM	File folder	

Name	Date modified	Type	Size
 3_Jim Cobb	2/18/2020 10:14 PM	File folder	
 4_Frank Hardeman	2/18/2020 10:15 PM	File folder	
 5_Palseo	2/18/2020 10:15 PM	File folder	
 6_Owen Jones	2/18/2020 10:15 PM	File folder	
 7_Jet Hicks	2/18/2020 10:15 PM	File folder	

Figure 1 – Different representations of folder Y containing m subfolders (Z)

But whatever naming criteria one adopts, each subfolder contains the articles that describe a specific event.

Name	Date modified	Type	Size
 Charlotte Daily Observer_09-23-1906_1_1.txt	2/8/2020 2:42 PM	TXT File	6 KB
 Chicago Daily Tribune_09-23-1906_1_5.txt	2/8/2020 2:42 PM	TXT File	7 KB
 Chicago Daily Tribune_09-24-1906_8_4.txt	2/8/2020 2:42 PM	TXT File	3 KB
 Daily Press_09-23-1906_1_1.txt	2/8/2020 2:42 PM	TXT File	5 KB
 Dallas Morning News_09-23-1906_2_1.txt	2/8/2020 2:42 PM	TXT File	1 KB
 Dallas Morning News_09-23-1906_2_1_2.txt	2/8/2020 2:42 PM	TXT File	1 KB
 Dallas Morning News_09-23-1906_2_1_3.txt	2/8/2020 2:42 PM	TXT File	1 KB
 Dallas Morning News_09-23-1906_2_1_4.txt	2/8/2020 2:42 PM	TXT File	1 KB
 Dallas Morning News_09-23-1906_2_1_5.txt	2/8/2020 2:42 PM	TXT File	1 KB
 Dallas Morning News_09-23-1906_2_1_6.txt	2/8/2020 2:42 PM	TXT File	1 KB
 Los Angeles Time_09-23-1906_2_4.txt	2/8/2020 2:42 PM	TXT File	4 KB
 Los Angeles Times_09-24-1906_1_2.txt	2/8/2020 2:42 PM	TXT File	8 KB
 Los Angeles Times_09-24-1906_6_1.txt	2/8/2020 2:42 PM	TXT File	1 KB
 New York Times_09-23-1906_1_1.txt	2/8/2020 2:42 PM	TXT File	9 KB
 New York Times_9-24-1906_2_5.txt	2/8/2020 2:42 PM	TXT File	11 KB

Figure 2 –Representation of folder X with n unsorted files

The way the algorithm works

The tool runs the NLP package you selected in the NLP Suite setup (Stanford CoreNLP, Stanza, or spaCy) with the NER annotator on each of the n documents in folder X and constructs an index of similarity based on social actors and NER values. It then computes a summary index of group values (based on social actors and NER values) for the files contained in each Z (m of them) we construct a summary index. Once the indices are computed, the algorithm compares the similarity value of each of the n file in X with the index of each Z subfolder to see if the processed document in X belongs to group Z.